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Imagining Open-Source Ecosystems for Experimental Weaving: a View from the AdaCAD Project

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This paper describes and reflects on a series of activities, performed from 2024-present, that looked to grow an open-source ecosystem centered around AdaCAD, a tool for what we began to call experimental weaving. We describe how we came to see experimental weaving as a term that can capture approaches and applications for weaving beyond common characterizations that center disciplinary affiliation or professional identity. We sketch the boundaries and sites of experimental weaving as it relates to our project and offer insights on common needs and challenges our project presented for these groups. We conclude by envisioning broader ecosystems for experimental weaving that can honor the specificity and diversity of practices that the term brings together

Keywords: experimental weaving; open-source; design software; TI-2026.

1. Introduction

Across the humanities and sciences, there is a growing interest in weaving. Whether weaving is used in the context of reflecting on embodied material relationships (e.g. Fernaeus et al., 2012; Oogjes & Wakkary, 2022); investigating cultural legacies of computing systems (e.g. Rosner, 2018); or building so-called "smart, soft and deformable structures (Buso et al., 2023; Kilic Afsar et al., 2021); weaving subcultures bubble up and trouble established categories of art, craft, design and engineering. AdaCAD, available online at www.adacad.org, is a computer aided design (e.g. CAD) system we designed to fuse computational design approaches common within engineering with the activities of weave drafting. Specifically, AdaCAD is a system that uses parametric design logics (e.g. creating designs by composing "dataflows" of functional units) as a weaving notation for the creation and

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documentation of weave structures (Devendorf et al., 2023). The project emerged from Devendorf's experience learning drafting and weaving. Namely, as a person who studied computer science, she found herself understanding and composing weave drafts via the algorithmic logics of computer science. For example, she saw a twill draft as a sequence of values that shifts over time according to specific rules (e.g. each row shifts only one unit in the left or right direction). AdaCAD implemented this logic with the mindset that it could support others who approached draft making as a form of "computational thinking" (Wing, 2006). In 2024 we received a grant from the United States National Science Foundation to conduct a series of planning activities that assessed if, and how, we might be able to grow an "open-source ecosystem" around AdaCAD (*POSE*, 2024). A software resource is open-source if it adheres to specific rules around access, use and re-use of its underlying source code ("The Open Source Definition," 2026). Non-code resources can be labeled as "open" via creative commons licenses that assert if and how the resources can be used and extended by a distributed community ("Creative Commons," 2026). An open-source ecosystem describes the socio-technical infrastructures that develop around a particular set of resources to ensure their availability and value to the community. Our grant looked to establish the basis of a thriving ecosystem with a robust network of users and contributors; a long-term plan for sustainability; and the ability to make new science/technology/art possible. We took projects like Scratch (Resnick et al., 2009) and P5.js (McCarthy, 2013) as exemplars of projects that had governance structures, user interfaces and resources that supported thriving open-source ecosystems and considered how these communities might take shape in the context of weaving.

As part of this project, we conducted interviews, hosted workshops, rewrote our documentation and continued to present the project at related venues. We also hosted a three-day gathering with 10 key stakeholders in weaving and/or open-source to reflect on our project and plan next steps. Since the start of the grant, we have seen our user base grow by 300% and have also onboarded several new project contributors. We reflect on what we learned through these activities with an eye towards aspects of our experience that would be relevant to the Textile Intersections community more broadly. First, we share how we have come to see the members of an AdaCAD ecosystem as "experimental weavers." We describe how we see this term and reflect on the benefits it has offered us in practice. Next, we focus on how our software has been received, used and not used so that others looking to develop novel resources can consider how similar factors might affect their process. Third, we synthesize our experiences to identify key opportunities and considerations for open-source in experimental weaving. While we are not the first to map and reflect upon transdisciplinary textile practices (Kilford et al., 2023; Pouta & Mikkonen, 2022), our project centers on the role open-source ecosystems might play in facilitating collaboration.

2. Research Context and Activities

The AdaCAD software has been released and continuously updated since 2018. During the first several years we focused on building out and improving the software. From 2024-2025 we used the NSF POSE grant to broaden our focus to include community building efforts.

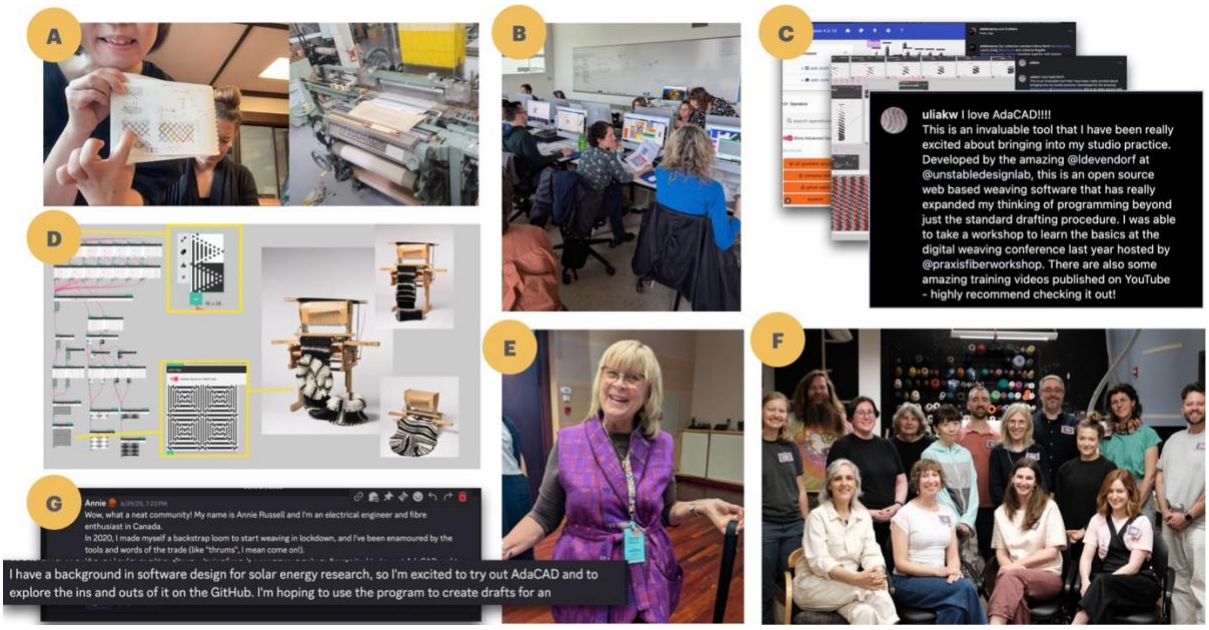


Figure 1: Summarizing Outreach Activities: (A) Kaori Ueda explains how she translated the multi-dimensional woven structure to work with Nishijin looms used for Kimono production. (B) A demonstration of AdaCAD to ~50 educators at the Midwest Fiber Arts Educators Network meeting in Detroit, MI in 2024. (C) Users show their workspaces and express gratitude for AdaCAD on Instagram (D) Berlin-based mechanical engineer turned weaver Ineyong Song shares how she used AdaCAD to make shape changing structures on a fully manual loom. (E) Laurie Carlson Steger, aka the LightWeaver, models a vest made from cloth she designed in AdaCAD (F) Leaders in weaving and creative coding across academia and industry met in June 2025 to develop strategy for AdaCAD (G) While introducing themselves via Discord, a new AdaCAD user highlights the unique intellectual backgrounds, which is a characteristic of many members of our community.

2.1 Positionality and Process

The authors on this publication represent the current leadership of the AdaCAD project: Laura Devendorf is an Associate Professor at CU Boulder and acts as the project lead; Deanna Gelosi is an AdaCAD developer and PhD Candidate focusing on creative computational design; and Etta Sandry is an outreach coordinator and PhD Candidate focusing on sample making and community formation in weaving. Former PhD students have also had strong impact on the AdaCAD project, namely Shanel Wu (Wu et al., 2024; Wu & Devendorf, 2020) and Mikhaila Friske (Friske et al., 2019). The project emerged in an academic setting, which has led us to focus on securing funding via large national science grants and sharing outcomes in peer-reviewed design research and engineering publications. While Etta has formal education in fiber arts at undergraduate and graduate levels, Laura is a self-taught weaver who tends to focus on weaving complex structures and Deanna is beginning weaver. While we use a variety of western-style looms, including dobby looms, we mostly commonly use a TC2 digital loom. Our access to a jacquard loom has shaped AdaCAD in workflows and supports for the development of complex and non-repeating structures. The woftware has also been shaped by collaborations cultivated through our experimental weaving residency program (Devendorf et al., 2020).

2.2 Eco-System Focused Activities

At the start of the grant term, AdaCAD consisted of the online user interface, a series of YouTube videos showing different uses for the software and limited documentation about the interface. We had roughly 100 registered users, which is a metric that tracks how many people “create an account” with AdaCAD, which allows them the ability to save their work. Contributions largely took the form of new code features and were produced in close collaboration with the project team.

In Fall 2024, We joined a cohort of grant recipients in a seven-week business model development course (NSF, 2027). During this process, we conducted 60 interviews with various stakeholders in our project to identify key "value propositions" (Eakin, 2024) that AdaCAD offered to its community. We recruited these participants from our local network of people who either already use AdaCAD, already experiment with weaving in some capacity but didn't use AdaCAD and people who develop other software for craft contexts. Additionally, we posted an open call for participants on our Instagram channel and received volunteers with whom we had not previously interacted. We approached each 30-minute virtual interview with a semi-structured set of prompts about their practice, their relationship with weaving and/or drafting software and how they might like to participate in an open-source community. We took notes during each interview and would discuss for 10 minutes post-interview to discuss how we thought the interview related to a list of value propositions that we continuously refined and shared with our cohort each week. We unpack some of these findings in Section 4.

After the interviews we focused on updating the software and our documentation resources based on what we had learned. We released an updated documentation website at docs.adacad.org, a new version of the software and launched a Discord community in January 2025. We celebrated this release with series of online workshops and office hours devoted to teaching the software to new communities. Our findings from these activities are anecdotal, as we had no systematic method for assessing workshops, office hours and Discord engagements. Next, we invited 10 experts in weaving and open-source to participate in a 3-day gathering devoted to shaping the future-plans for the project and ecosystem. This included presentations of the software, in-person workshops exploring the software on different looms and brainstorming sessions devoted to possible features. We took detailed notes and photographs of these events and reflected on them through an internal working document that described what we did and learned and how the event might shape what we do next. Lastly, we traveled to 3 different community based and/or academic research labs to conduct in-person AdaCAD training workshops. These workshops were assessed informally, through reflections on the part of the workshop leaders and feedback volunteered by participants.

At the end of the Summer 2025 our number of registered users had grown to 400 and we began to see more varied contributions to AdaCAD, including code updates, the production of informational resources and instances of members teaching AdaCAD to others. While it is difficult to quantify and compare these different forms of contribution, we have heard reports of at least five people independently teaching AdaCAD to their local community members, we observe at least three members on our Discord answering questions posted by others and have had three developers invest time in offering improvements and additions to the project software.

3. Defining Experimental Weaving

The business model development course pushed us to clarify who counted as a "user" or "contributor" and the value AdaCAD delivered for these groups. We struggled to find a single descriptive term that captured the breadth of practices we had in mind. There were "complex weavers"(Complex Weavers Guild, 2012) looking to push the boundaries of weaving; research labs in engineering looking for ways to add textiles to their research programs; and educators who may not use AdaCAD but were inclined to teach it to students. At the same time, we faced challenges in making weaving legible and relevant

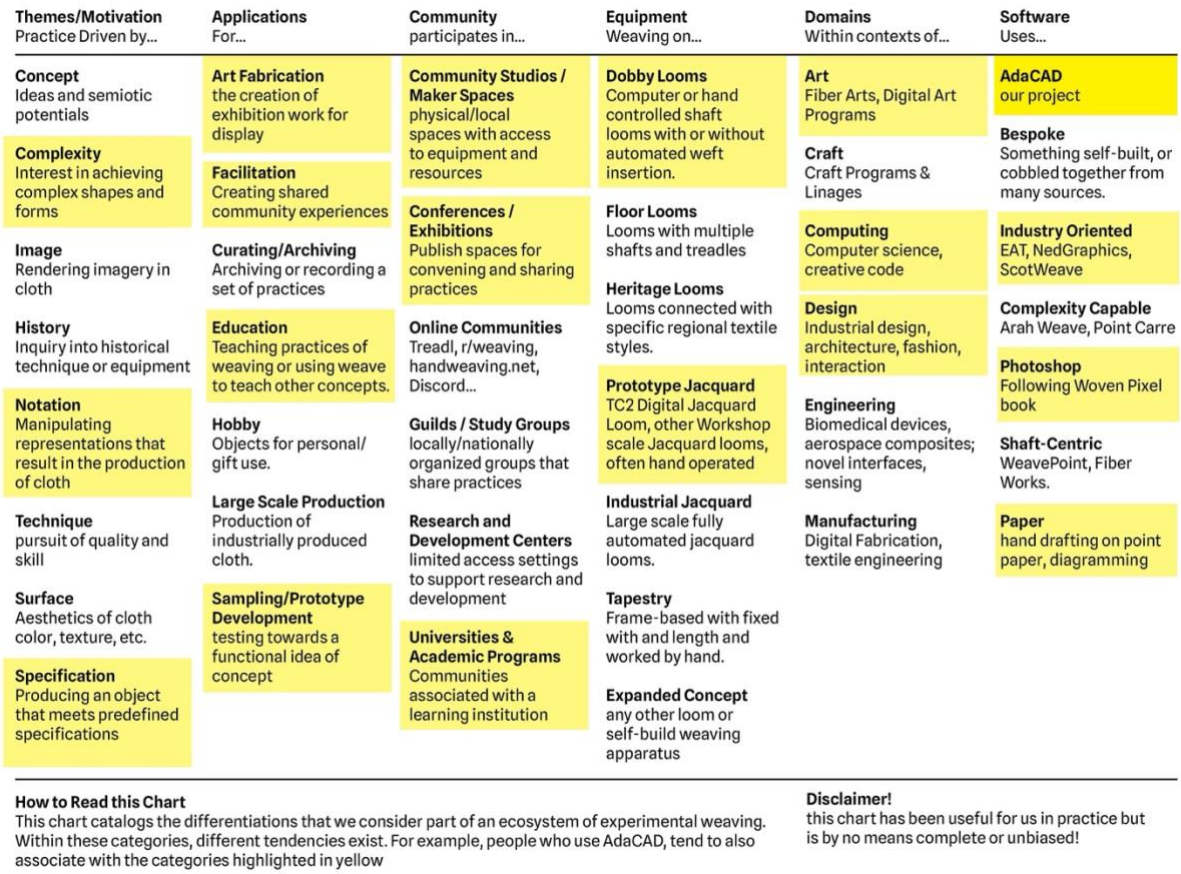


Figure 2 offers a map of where we have seen experimental weaving growing in and around AdaCAD. We list this emergent set of categories to support others in surveying this landscape while acknowledging that our list is partial and non-exhaustive.

to other course participants and business. They tended to read it as a "hobbyist" practice or fashion-adjacent. Their attitude seemed to shift when we demonstrated how textiles and weaving are active methods of inquiry in biomedical applications and aerospace composite materials development and that those fields stand to benefit from the expertise of so-called "hobbyists." These conversations crystallized a frustration we had carried for a while: standard categorizations don't capture our project or its participants. Experimental weaving is a term we had already been using as part of an artist residency program (Unstable Design Lab, 2026a) and talk series (Unstable Design Lab, 2026b) with the intent of identifying practices by their approach and ethos, as opposed to disciplinary domain. We started using it in our course discussions as a means of: (1) interrupting snap judgements that come with the association of particular materials, lifestyles and accreditations; and (2) opening pathways for new communities under this action-centered definition – as if to say, "if you consider yourself someone who experiments with weaving in some capacity, you are welcome here." Seeing experimental weaving as a context for AdaCAD helped us situate our project as one of many possible interventions. Put another way, we could cultivate the community alongside, or even in absence, of the software, making us less reliant on the success of this intervention in our broader goals of elevating and equipping experimental weavers with resources.

4. Challenges and Opportunities in Designing for Experimental Weavers

Under this ecosystem of experimental weaving, AdaCAD tends to be valued by practitioners who share one of two tendencies: First, there are people who already had a weaving practice and/or had received training in weaving. Second there are "weave curious" people are relative outsiders to weaving but are becoming interested in the potentials it might offer to their practices.

4.1. For Established Weavers, Learning AdaCAD Resembled Learning a New Instrument

Many weavers with established drafting approaches find our software's parametric approach to be a barrier to entry. They can interact with it and see its value, but if their existing workflow is working for them, they are unlikely to invest time in understanding a new approach and way of thinking. This leads us to see drafting as a process that is tightly coupled to existing workflows and software systems. For example, using Photoshop for drafting is not just a means to an end, it fundamentally structures how the cloth is conceptualized. We understand this experience akin to learning a new instrument – if weavers are adept pianists, our software appears as a flute. Both offer interesting potentials, but the overhead of unlearning one system to learn another can feel daunting.

However, even if established weavers are hesitant to learn AdaCAD for their own practice, they expressed enthusiasm for integrating AdaCAD into teaching activities. They feel it could resonate with a new generation of students who may have already encountered flow languages and computational thinking principles in school. They saw the online nature of the software as something that could smooth friction when working in settings that offer varied looms. Additionally, the software is also somewhat loom agnostic, allowing designs to be adapted across available course looms.

For others with existing practices, the parametric style offered by AdaCAD matches their internal modes of thinking in a way that other methods have not yet captured. These participants feel "seen" and energized by the process and quickly invest in learning the different abilities. Others find the challenge of relearning or rethinking their process to be part of the value. This is most frequently seen with retired weavers, who find the challenge of relearning to be enjoyable and creatively stimulating. These weavers were also the most involved in virtual workshops and Discord contributions.

One powerful motivator for established weavers to invest in learning AdaCAD is the experience of being "locked out" of advanced and/or expensive software they had access to through their educational programs. AdaCAD offers an avenue for participation in complex draft making outside well-resourced departments. In parallel, programs like LMRM[1] and Praxis[2] in the United States are emerging to provide access to advanced weaving machinery outside of academic institutions, creating additional need for accessible complex drafting software.

In response to these findings, we focused on developing resources to support the learnability of our software. We wrote explanations of parametric design, created step-by-step tutorials and designed worksheets that could slot into existing course curriculum. We focused on common teaching tasks like weaving imagery, sometimes directly contrasting AdaCAD's approach with other popular methods. For example, in Figure 3 we show how we contrast our approach to image-based with the one Figure 3: Examples of the kinds of materials we generated to support established weavers understand

[1] https://www.lmrmchicago.com/
[2] https://www.praxisfiberworkshop.org/

AdaCAD. Left, we describe parametric operations in our software in material comparison to a guitar pedal—an operation manipulates drafts like a guitar pedal manipulates sound. Right, we show two approaches to image-based weaving—one using AdaCAD and the other using the commonly used methods from The Woven Pixel.

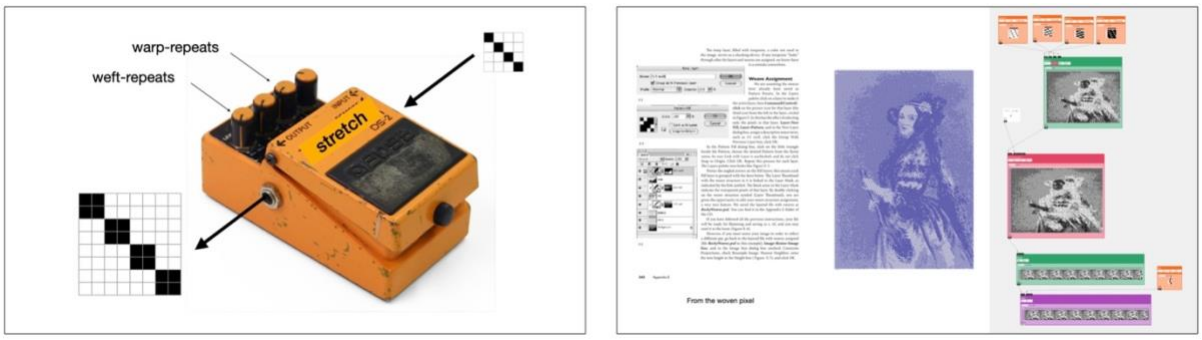


Figure 3: Examples of materials we generated to support established weavers in understanding AdaCAD. Left, we describe parametric operations in our software using guitar pedals as an analogy—an operation manipulates drafts like a guitar pedal manipulates sound. Right, we show two approaches to image-based weaving—one using AdaCAD and the other using the commonly used methods from The Woven Pixel.

developed in The Woven Pixel (Schlein & Ziek, 2006). We also added workspace sharing and weaving information file (.WIF)[3] exports to support computer-controlled dobby looms and interoperability with other weaving software. Collectively, we saw these updates as an effort to meet people where they were at, helping them better understand our approach for both personal explorations and integration into teaching.

4.2. Helping Weave Curious People Interpret and Act with Drafts

We observed several ways that AdaCAD aided in the understanding and extensions of a weaving practice for weave curious people. Among these users, parametric design tends to be more familiar than weaving and drafting. They tend to see the draft as a black and white graphic – rather than an abstraction of machine instructions. For some, this is an invitation to rethink the context of AdaCAD, for instance, by using it to create other kinds of stitching patterns rather than weaving patterns specifically. Others experiment by creating visually interesting drafts, only to be disappointed when the cloth produced does not have the same visual qualities, or, when the draft is not feasible for their loom to support. In response, we developed examples that treat the draft as a "graphic" to realize in a cloth through ready-made conversions using overshot or damask techniques. We also created loom-first tutorials, like how to use AdaCAD to create threading and treadling plans for multi-shaft loom configurations. We also began integrating more material translations into our workshops to help weave curious people quickly understand how a draft translates into a material. Specifically, we would begin workshops with examples of complex forms and notations already existing within weaving (e.g. establishing legitimacy/potential) and then invite hands-on practice by weaving structures with paper.

[3] The .WIF specification is a standard format for weaving described at (Nielsen, 1997)

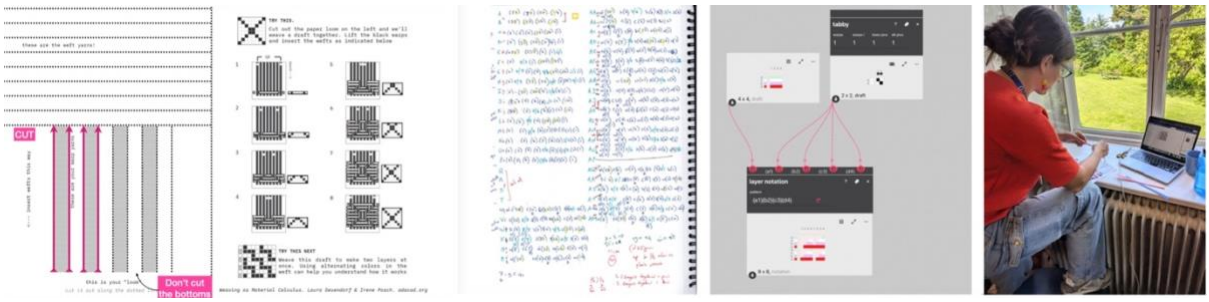


Figure 4: resources we developed to support weave curious participants. From left to right: a paper loom and instructions for how to interpret drafts. Next, a side-by-side translation of Kathryn Walter's bespoke weaving notation and its equivalent in AdaCAD. A participant hand weaves a draft they created with paper.

In three cases, participants in our workshops who identify as weave curious engaged in software development aspects of the project by writing their own custom operations or adding features that aid in AdaCAD's broader usability. More broadly, people with coding skillsets tend to be quick to play and interested in sharing their outcomes with a more diverse community than typical open-source projects. They also take more readily to our GitHub and Discord communities, as we believe that this mode of communication is more established among these groups. We also began using GitHub's issue tracking feature to help potential contributors identify key areas where we needed support and had some success with contributors completing these tasks. These experiences demonstrate that an experimental weaving community can present interesting puzzles, tasks and opportunities for weave curious computer programmers.

Several established weavers noted how AdaCAD's real time updates (e.g. changing a parameter immediately changes the draft) made weaving conventions "explainable" to their weave curious collaborators. For example, in emerging engineering contexts where teams collaborate with or hire experienced weavers for prototype development, our system became a way to convey design constraints and conventions to non-weavers. Another weaving participant recalled frustration in her desire to collaborate with engineers because they would not take her seriously. She felt like the presentation AdaCAD offered of weave drafting as an algorithmic process would have helped in these settings to bring legitimacy and relevance to her practice. These anecdotes demonstrate the communicative potentials of design software and how it can serve as a boundary object (Star & Griesemer, 1989) for the various sub-groups of experimental weaving.

5. Discussion

Our use of the term experimental weaving and our interventions with AdaCAD required us to envision and enact a nascent community that cuts across established definitions and prejudices associated with different practices. At the same time, we had to resolve real differences in user needs: some users are so well versed in weaving that taking on new mode of working is costly; and some users so un-versed in weaving that they face difficulty in understanding the abstractions used for weaving. Between these poles sit interesting variations, like the creative coders who delight in contributing their own functions inspired by innovations in aligned fields, retired weavers testing the software's limits and mentoring others. The takeaway, perhaps, is that neither weavers nor engineers are monoliths. Rather than bringing "new tech" to "old weaving", we are attempting to dissolve these categorizations so that other associations, commonalities and practices can grow.

5.1 Potentials for Open-Source Experimental Weaving Communities

Weavers are resourceful and our experiences revealed how weavers co-invent their tools and systems. The notations that weavers use in designing cloth can be highly individualized, working across paper, spreadsheets, hand-drawn diagrams and software. Sometimes weavers develop bespoke systems to convert between varied or completely outdated technologies for controlling looms. In line with other researchers (Orynek et al., 2023), we see real opportunities for an open-source ecosystem to cater to these bespoke aspects of weaving. An open-source mindset, in which tools with many variations and modes of representation are shared for public use, derivation and non-commercial intent, gives such a community pathways to exchange and collaborate. We aim for AdaCAD to be a seed that can give rise to a larger ecosystem for creative collaboration in weaving. We share our experiences in a hopeful gesture to support others who might aim situate themselves in an experimental weaving ecosystem or build their own ecosystems. At the same time, we are mindful that software, alone, does not sustain community. While our focus has been on the people circulating around our project, we are actively exploring governance models, such as one defined as "circles" (Schneider & Hornbein, 2026), to structure distributed collaboration and bring more voices to the table to shape the future of the project.

5.2 Rethinking Success

The varied and nascent nature of an experimental weaving community also creates challenges in traditional measures of success. In conversation with leaders of other arts-centered open-source communities, we have heard stories of software that "took off," and we expected that if AdaCAD was valuable to others, its user base and online activity would rapidly grow too. Other groups in our business development course cohort supported millions of users and our business mentors in the course continually emphasized growth as a metric for success. At the end of our outreach efforts, we had about 400 registered users and it's unclear how many of those people used AdaCAD more than once. This is not a large number by open-source standards, but our project also doesn't look like a typical open-source project. For example, many of the projects we compare ourselves to are for coding communities by coding communities, with shared norms about what it means to contribute and participate. Experimental weaving, by contrast, can have contributions that go beyond code. We are looking to Discord to support community exchange and coordination of non-code contributions. For instance, users can post or answer questions; share completed projects; advertise AdaCAD related events or highlight inspirational projects and images to others in the group.

Some contributions to experimental weaving also require access to specialized machinery which can add additional barriers to participation. Additionally, one does not need to design and weave a complex draft multiple times a day, so, one might expect the usage to be slower than, say, a software library built into various other projects. This may also be why establishing commercial markets for such tools tend toward expensive licensing—and more reason to develop venues for sharing or collectively building and maintaining tools instead. While open-source comes with its own political legacy (Coleman, 2004) and wrestles with its relationship to unpaid labor (Terranova, 2000), it has been a worthwhile endeavor for us as grant funded academic researchers to offer our labors in an open capacity.

For us, success has been registered most directly in the relationships that we built around the software – collaborations and relationships we may not have otherwise encountered and in the delight in seeing others find value and joy in the system, whether it takes the form of a hand-made vest, a master's

thesis (Nayar, 2024), or a nerdy and thrilling conversation about ideas in weaving and combinatorics (Devendorf et al., 2023). These few but deep engagements have helped us secure support from funding bodies, which in turn fed subsequent development, grant writing and publishing. Our incentives, then, are to create more avenues for experimental weaving to be recognized and valued, rather than to make sure our specific intervention dominates others in a free market—when relational gains are valued over financial ones, we see more room to cater to the strange, weird and esoteric rather than homogenizing practice into standardized workflows.

6. Conclusion

Through our work growing an open-source ecosystem around AdaCAD, we have come to see experimental weaving as a community that cuts across existing disciplinary and cultural categories. We believe an open-source ecosystem for experimental weaving can help establish collaborative relationships and symbiotic partnerships between typically siloed groups. At the same time, it invites reflection on shared norms and practices around contribution and collective authorship, on infrastructures for archiving and teaching, as well as on a willingness to honor practices that engage different values, bodies, and machines.

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